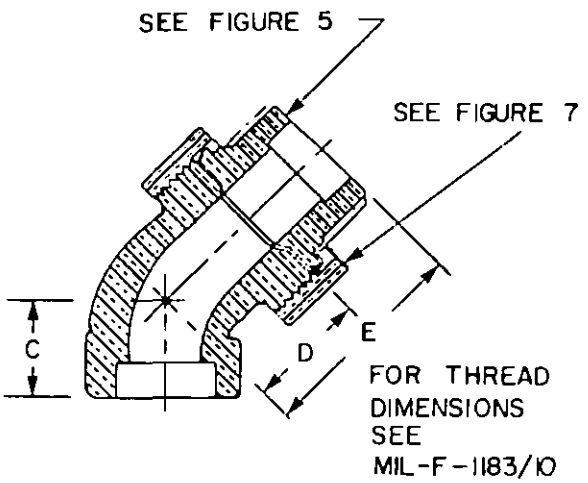
MILITARY SPECIFICATION SHEET

FITTINGS, PIPE, CAST BRONZE, SILVER-BRAZING; UNION ELBOW, PIPE AND UNION TEE, PIPE

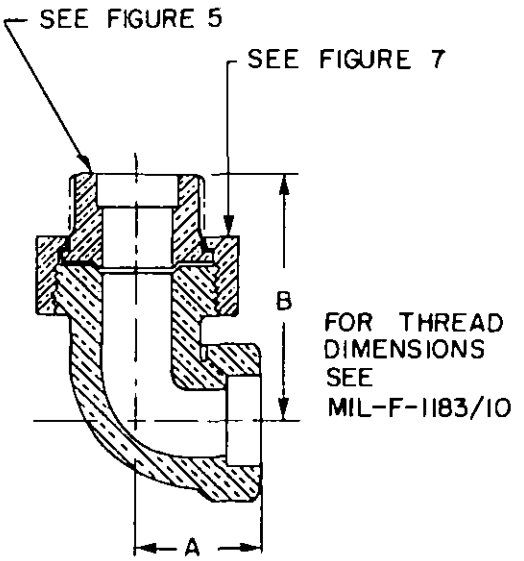
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the union elbows and union tees described herein shall consist of this specification and the latest issue of MIL-F-1183.



SH 12192

FIGURE 1. 45-degree union elbow.



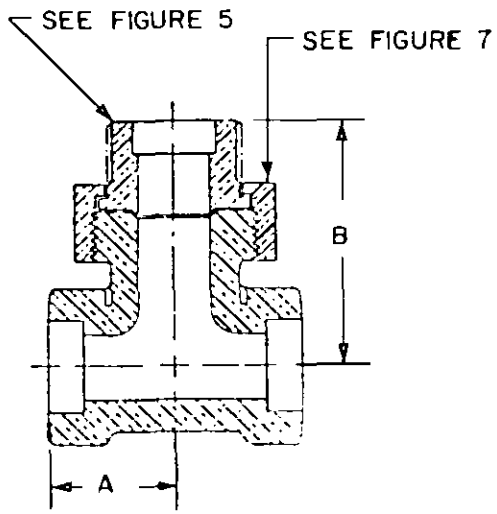
SH 12193

FIGURE 2. 90-degree union elbow.

AMSC N/A FSC 4730

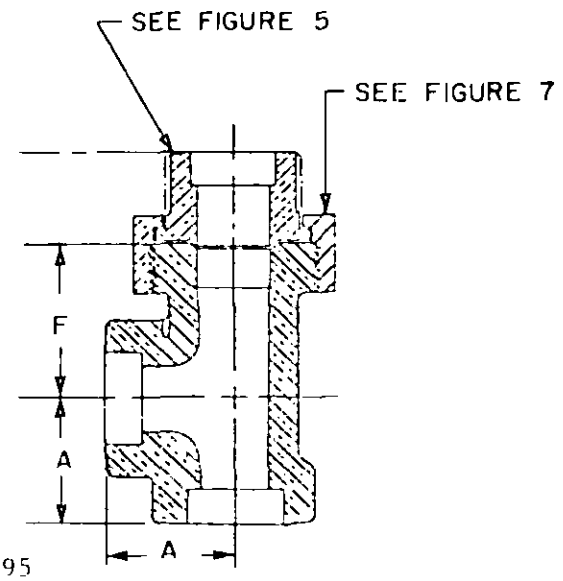
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SH 12194

FIGURE 3. Union branch tee.

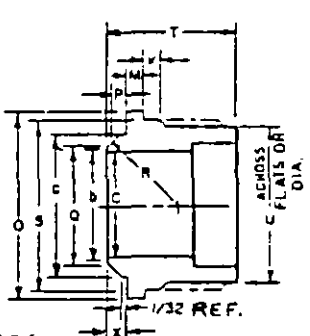


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FIGURE 4. Union run tee.

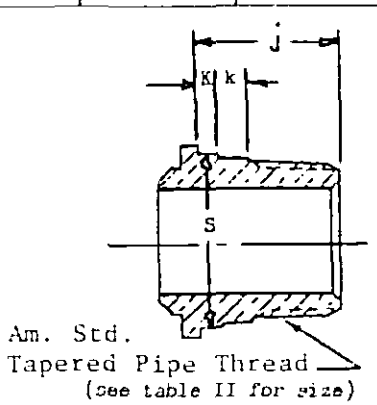
TABLE I. Center-to-end dimensions.

Nominal pipe size (Inches)	A (Inches)	B (Inches)	C (Inches)	D (Inches)	E (Inches)	F (Inches)
1/4	15/16	1-15/16	13/16	25/32	1- 9/16	1- 5/32
3/8	1- 1/16	2- 7/32	7/8	55/64	1- 3/4	1-11/32
1/2	1- 1/4	2-13/32	1	1	1-15/16	1-15/32
3/4	1- 7/16	2- 7/8	1- 1/8	1- 1/8	2- 3/16	1-13/16
1	1- 5/8	3- 5/32	1- 5/16	1-19/64	2- 1/2	1-31/32
1-1/4	1-15/16	3-21/32	1- 1/2	1- 1/2	2- 7/8	2-19/64
1-1/2	2- 1/8	3-15/16	1-11/16	1-21/32	3- 1/8	2- 7/16
2	2- 1/2	4- 1/2	2	1-15/16	3-11/16	2- 3/4



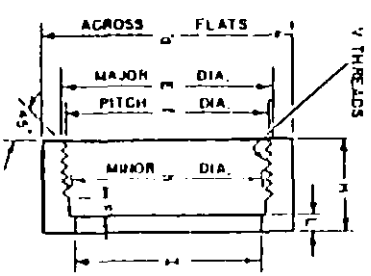
SH 12196

FIGURE 5. Female tailpiece.



SH 12197

FIGURE 6. Male thread tailpiece.



SH 12198

FIGURE 7. Nut.

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TABLE II. Union components dimensions.

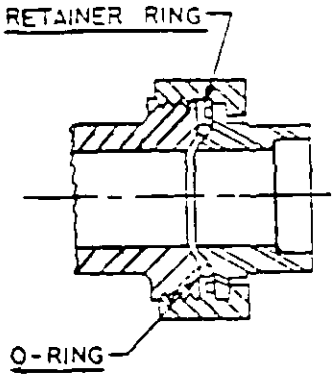
Nominal pipe size	C Diameter		E	F ± .016	G	H Min	I Diameter	K	L Min	M ± .010
Inches	Inches	±	Inches	Inch	Inches	Inches	Inches	Inch	Inch	Inch
1/4	0.398	0.015	1.095	0.120	1-11/32	21/32	0.860	0.125	0.125	0.125
3/8	.532	.015	1.281	.125	1-35/64	45/64	1.020	.135	.135	.135
1/2	.697	.020	1.491	.135	1-25/32	47/64	1.220	.145	.145	.145
3/4	.907	.025	1.835	.150	2- 3/16	7/8	1.510	.160	.160	.160
1	1.171	.025	2.165	.170	2-35/64	59/64	1.820	.175	.175	.180
1-1/4	1.502	.030	2.601	.200	3- 3/64	1-1/32	2.220	.205	.205	.210
1-1/2	1.742	.030	2.921	.220	3-25/64	1-1/16	2.520	.220	.220	.230
2	2.186	.030	3.521	.250	4- 3/64	1-5/32	3.090	.250	.250	.260

Nominal pipe size	O Diameter	P	Q	R Radius	S Diameter	T ± .020	U	V	X	Pipe thread size
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches		Inch	Inches
1/4	0.965	0.100	0.510	0.361	0.83	0.874	0.75	16-UNS	0.134	3/8
3/8	1.135	.103	.650	.460	.99	.959	.90	14-UNS	.139	1/2
1/2	1.345	.113	.835	.591	1.19	1.049	1.09	14-UNS	.159	3/4
3/4	1.650	.115	1.030	.728	1.47	1.154	1.35	11-UNS	.154	1
1	1.975	.135	1.315	.930	1.78	1.299	1.63	11-UNS	.184	1-1/4
1-1/4	2.395	.130	1.650	1.167	2.18	1.469	2.00	10-UNS	.174	1-1/2
1-1/2	2.710	.143	1.915	1.355	2.47	1.619	2.27	10-UNS	.199	2
2	3.310	.165	2.395	1.694	3.04	1.854	2.80	10-UNS	.224	2-1/2

Nominal pipe size	b Diameter	c Diameter	f Pitch diameter		h Minor diameter		j	k
			Minimum	Maximum	Minimum	Maximum		
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
1/4	0.441	0.618	1.0520	1.0585	1.0278	1.0348	0.985	0.30
3/8	.576	.768	1.2320	1.2390	1.2043	1.2122	1.155	.30
1/2	.741	.972	1.4420	1.4492	1.4111	1.4190	1.165	.30
3/4	.951	1.174	1.7720	1.7801	1.7326	1.7424	1.350	.31
1	1.216	1.494	2.1020	2.1102	2.0626	2.0724	1.505	.33
1-1/4	1.561	1.828	2.5321	2.5407	2.4887	2.4992	1.535	.33
1-1/2	1.801	2.116	2.8520	2.8609	2.8087	2.8192	1.610	.39
2	2.276	2.638	3.4520	3.4610	3.4087	3.4192	2.020	.39

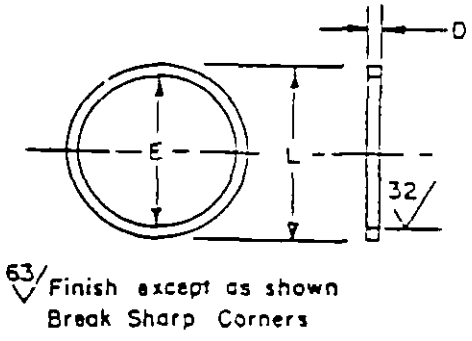
NOTE: For socket details see figure 1 and table V of MIL-F-1183.

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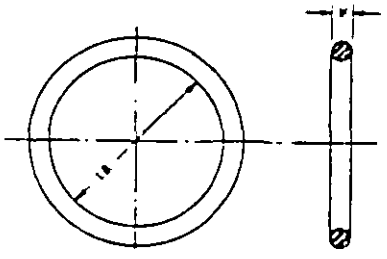
SH 12199

FIGURE 8. O-ring and retainer ring placement.



SH 12200

FIGURE 9. Retainer ring.



SH 12201

FIGURE 10. O-ring.

TABLE III. Retainer ring dimensions.

Nominal pipe size	D ± .005 (Inch)	E ± .005 (Inches)	L ± .005 (Inches)
1/4	0.050	0.813	1.008
3/8	.050	.938	1.184
1/2	.050	1.188	1.394
3/4	.060	1.500	1.715
1	.070	1.750	2.045
1-1/4	.070	2.062	2.471
1-1/2	.070	2.375	2.791
2	.090	3.000	3.391

TABLE IV. O-ring configuration and size.

Nominal pipe size	O-ring configuration and size
1/4	AS568-017
3/8	AS568-019
1/2	AS568-023
3/4	AS568-125
1	AS568-129
1-1/4	AS568-134
1-1/2	AS568-139
2	AS568-232

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REQUIREMENTS:

1. Female union elbow type. Forty-five degree and 90-degree union elbows with female tailpiece (see figures 1 and 2) shall be type A (streamlined interior).
2. Male union elbow type. Forty-five degree and 90-degree union elbows with male tailpiece shall be type A (streamlined interior).
3. Female union tee type. Union branch and union run tees with male tailpiece (see figures 3 and 4) shall be type A (streamlined interior).
4. Male union tee type. Union branch and union run tees with male tailpiece shall be type A (streamlined interior).
5. Center-to-end dimensions. The center-to-end dimensions for union elbows and union tees shall be as specified in table I.
6. Female tailpiece dimensions. Dimensions for the female tailpiece (see figure 5) shall be as specified in table II.
7. Male tailpiece dimensions. Dimensions for the threaded male tailpiece (see figure 6) shall be as specified in table II. Male tailpiece dimensions are the same as the dimensions for the female tailpiece (see figure 5) except for those dimensions identified on figure 6. On figure 5, the dimensions P, Q and R define the point of contact of the ground seat. NOTE: The size of the male pipe thread is one size larger than the nominal pipe size by which the union elbow or union tee is identified.
8. Nut dimensions. Dimensions for the union nut (see figure 7) shall be as specified in table II.
9. Static O-ring seal and retainer ring. When specified (see requirement 12), union elbows and union tees shall be furnished with an O-ring and a retainer ring which when installed is outboard of the ground joint (see figure 8). The O-ring and retainer ring do not affect the primary contact of the ground seat and forms an effective secondary seal. NOTE: For Navy applications, union elbows and union tees cannot be used without O-ring and retainer ring.
10. O-ring seal configuration and material. O-ring configuration shall be in accordance with AS 568 of size specified in table IV for the required union elbow or union tee nominal pipe size. O-ring material shall be MIL-R-83248 class 1 rubber.
11. Retainer ring dimensions and material. Retainer ring dimensions shall be as specified in table III. Material of retainer ring shall be QQ-C-390, alloy C90300 or C92200 or ASTM B 61 (UNS no. C92200) bronze.

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12. Specification sheet part number. Union elbows and union tees covered by this specification sheet shall be designated in the following form (see 6.2(b) of MIL-F-1183).

M1183/11 - XX X X

Specification sheet code number
Union-end fitting and pipe size code number (see requirement 13 herein)
Brazing ring groove and brazing ring code letter (see requirement 14 herein)
O-ring and retainer ring code number (see requirement 15 herein)

13. Union-end fitting and pipe size. The union-end fitting (45-degree union elbow, 45-degree male union elbow, 90-degree union elbow, 90-degree male union elbow, union branch tee, male union branch tee, union run tee or male union run tee) and the nominal pipe sizes are identified by two digits (see table V).

TABLE V. Code number to union-end fitting and pipe size.

45-degree union elbow	45-degree male union elbow	90-degree union elbow	90-degree male union elbow	Union branch tee	Male union branch tee	Union run tee	Male union run tee	Nominal pipe size
11	21	31	41	51	61	71	81	1/4
12	22	32	42	52	62	72	82	3/8
13	23	33	43	53	63	73	83	1/2
14	24	34	44	54	64	74	84	3/4
15	25	35	45	55	65	75	85	1
16	26	36	46	56	66	76	86	1-1/4
17	27	37	47	57	67	77	87	1-1/2
18	28	38	48	58	68	78	88	2

14. Brazing ring groove and brazing ring. Face fed (no groove), grooved without brazing ring, grooved with brazing ring inserted in groove, grooved with brazing rings separately packaged and attached to the union-end fitting or grooved with brazing rings separately packaged and shipped with but not attached to the union-end fitting socket ends are identified by a single letter (see table VI).

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TABLE VI. Code letter to socket condition.

Code letter	Condition of socket
N	No groove (face fed)
G 1/	Grooved (no brazing ring)
R 1/	Grooved with brazing ring inserted
A 2/	Grooved with brazing ring separately packaged and attached
S 1/	Grooved with brazing ring separately packaged and shipped with

1/ No National Stock Number (NSN) shall be established or exist in the Federal Supply System for a union-end fitting with code letter of G, R or S.

2/ When the socket condition code letter is "A", the brazing rings shall not be inserted in the grooves but shall be packaged separately and either included within each individual fitting package or physically attached to the fitting.

15. O-ring seal and retainer ring. Whether the union-end fitting is provided without O-ring and retainer ring, with O-ring and retainer ring installed, with O-ring and retainer ring separately packaged and attached to the union-end fitting or with O-ring and retainer ring separately packaged and shipped with the union-end fitting but not attached to the union-end fitting is identified by a digit (see table VII).

TABLE VII. Code number to O-ring and retainer ring requirement.

Code number	O-ring and retainer ring requirement
1/ 2	Without O-ring and retainer ring
3	With O-ring (air, oil or water service) and retainer ring installed
4	With O-ring (steam service) and retainer ring installed
1/ 5	With O-ring (air, oil or water service) and retainer ring separately packaged and attached
1/ 6	With O-ring (steam service) and retainer ring separately packaged and attached
1/ 7	With O-ring (air, oil and water service) and retainer ring separately packaged and shipped with
1/ 8	With O-ring (steam service) and retainer ring separately packaged and shipped with

1/ No National Stock Number (NSN) shall be established or exist in the Federal Supply System for a union-end fitting with code number of 2, 5, 6, 7 or 8.

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16. Specification part number definition. The specification sheet part number is a definitive part number which is formed by combining the specification sheet code number (M1183/11 for MIL-F-1183/11) with a dash after it, the union-end fitting and pipe size code number, the brazing ring groove and brazing ring code letter and the O-ring and retainer ring code number as specified in requirement 12.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activity:
DLA - CS

Preparing activity:
Navy - SH

Agent:
DLA - CS

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